

Evaluation of Recommender Systems

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Summary

Proper evaluation is important, but really difficult.

Evaluation: Typical Questions

- Do recommendations work? Do they increase sales? How much?
- Which algorithm should we prefer for our application?
- Which parameter setting is better?

Evaluation is Important

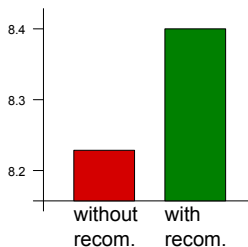
- many choices available: recommender techniques, similarity measures, parameter settings. . .
- personalization $\Rightarrow$ difficult testing
- impact on revenues may be high
- development is expensive
- intuition may be misleading

Evaluation is Difficult

- hypothetical examples
- illustrations of flaws in evaluation

Case I

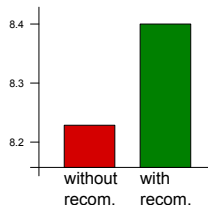
- personalized e-commerce system for selling foobars
- you are a manager
- I'm a developer responsible for recommendations
- this is my graph:



I did good work. I want bonus pay.

Case I: More Details

- personalized e-commerce system for selling foobars
- recommendations available, can be used without recommendations
- comparison:
 - group 1: users using recommendations
 - group 2: users not using recommendations
- measurement: number of visited pages
- result: $\text{mean}(\text{group 1}) > \text{mean}(\text{group 2})$
- conclusion: recommendations work!



really?

Issues

- what do we measure: number of pages vs sales
- division into groups: potentially biased (self-selection) vs randomized
- statistics: comparison of means is not sufficient
 - role of outliers in the computation of mean
 - statistical significance (p-value)
 - practical significance – effect size
- presentation: y axis

Case II

- two models for predicting ratings of foobars (1 to 5 stars)
- comparison on historical data
- metric for comparison: how often the model predicts the correct rating
- Model 1 has better score than Model 2
- conclusion: using Model 1 is better than using Model 2

probing questions?

potential flaws?

Issues

- over-fitting, train/test set division
- metric:
 - models usually give float; exact match not important
 - we care about the size of the error
- statistical issues again (significance of differences)
- better performance wrt metric $\Rightarrow$ better performance of the recommender system ?

General Evaluation Problem

- what we care about:
 - long-term sales
 - user satisfaction, trust, happiness, learning, ...
 - fairness, equity, diversity, ...
- what we can measure (and that's still non-trivial):
 - short-term sales
 - ratings, clicks, response times
 - predictive accuracy

Evaluation Methods

- experimental
 - “online experiments”, A/B testing
 - ideally “randomized controlled trial”
 - at least one variable manipulated, units randomly assigned
- non-experimental
 - “offline experiments”
 - historical data
- simulation experiments
 - simulated data, limited validity
 - “ground truth” known, good (not only) for “debugging”

Offline Experiments

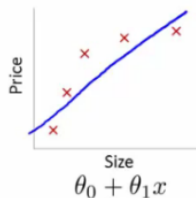
- data: “user, product, rating”
- overfitting, cross-validation
- performance of a model – difference between predicted and actual rating

predicted	actual
2.3	2
4.2	3
4.8	5
2.1	4
3.5	1
3.8	4

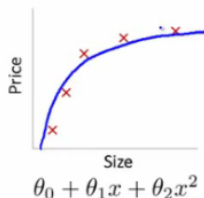
Overfitting

- model performance good on the data used to build it; poor generalization
- too many parameters
- model of random error (noise)
- typical illustration: polynomial regression

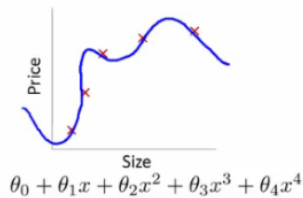
Overfitting – illustration



High bias
(underfit)



“Just right”



High variance
(overfit)

<http://kevinbinz.com/tag/overfitting/>

Cross-validation

- aim: avoid overfitting
- split data: training, testing set
- training set – setting model “parameters” (includes selection of fitting procedure, number of latent classes, and other choices)
- testing set – evaluation of performance
- (validation set)

(more details: machine learning)

Train and Test Set Division

typical setting (e.g., image classifiers):

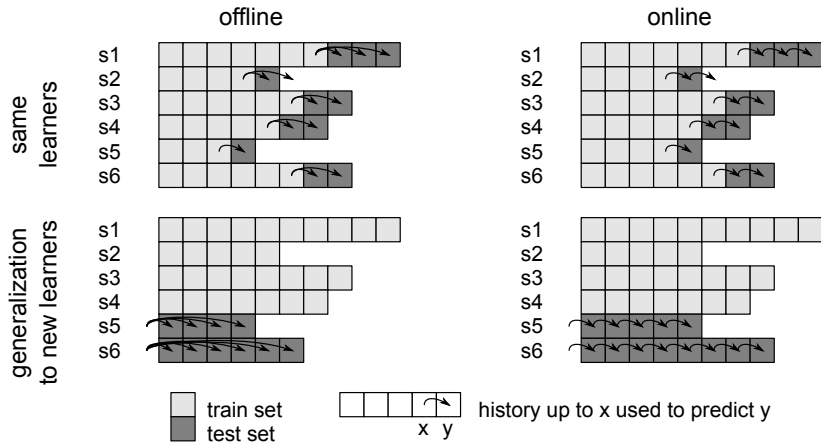
- 80 % train, 20 % test
- randomized selection
- k -fold cross validation: k folds, in each turn one fold is the testing set

Train and Test Set Division: RecSys

not so simple...

- data entries not independent, randomized selection not reasonable
- should the division respect user data? item data?
- temporal aspect: avoid “predicting past from future”, respecting time information
- k -fold cross validation while respecting time?

Train/Test Set Division



Bayesian Knowledge Tracing, Logistic Models, and Beyond: An Overview of Learner Modeling Techniques

Note on Experiments

- (unintentional) “cheating” is easier than you may think
- “data leakage”
 - training data corrupted by some additional information
- useful to separate test set as much as possible

Metrics

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4.8	5
2.1	4
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- MAE (mean absolute error)

$$MAE = \frac{1}{n} \sum_{i=1}^n |a_i - p_i|$$

- RMSE (root mean square error)

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (a_i - p_i)^2}$$

- correlation coefficient

higher is better? lower is better?

When do measures differ?

Describe specific cases of predictions and actual outcomes such that:

- ① good RMSE, good correlation
- ② good RMSE, poor correlation
- ③ poor RMSE, good correlation
- ④ poor RMSE, poor correlation

Normalization

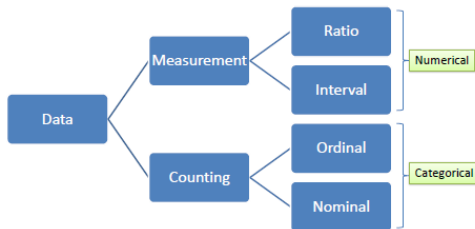
- used to improve interpretation of metrics
- e.g., normalized MAE

$$NMAE = \frac{MAE}{r_{max} - r_{min}}$$

Note on Likert Scale

1 to 5 “stars” ~ Likert scale (psychometrics)

what kind of data?



<http://www.saedsayad.com/data-preparation.htm>

Note on Likert Scale

- 1 to 5 “stars” ~ Likert scale (psychometrics)
strongly disagree, disagree, neutral, agree, strongly agree
- ordinal data? interval data?
- for ordinal data some operation (like computing averages) are not meaningful
- in RecSys commonly treated as interval data

Binary Predictions

- like
- click
- buy
- correct answer (educational systems)

prediction: probability p

notes:

- (bit surprisingly) more difficult to evaluate properly
- closely related to evaluation of models for weather forecasting (rain tomorrow?)

Metrics for Binary Predictions

- do not use:
 - MAE: it can be misleading (not a “proper score”)
 - correlation: harder to interpret
- reasonable metrics:
 - RMSE
 - log-likelihood

$$LL = \sum_{i=1}^n c_i \log(p_i) + (1 - c_i) \log(1 - p_i)$$

Information Retrieval Metrics

		Reality	
		Actually Good	Actually Bad
Prediction	Rated Good	True Positive (tp)	False Positive (fp)
	Rated Bad	False Negative (fn)	True Negative (tn)

All recommended items

All good items

- accuracy
- $\text{precision} = \frac{TP}{TP+FP}$
good items recommended / all recommendations
- $\text{recall} = \frac{TP}{TP+FN}$
good items recommended / all good items
- $F1 = \frac{2TP}{2TP+FP+FN}$
harmonic mean of precision and recall

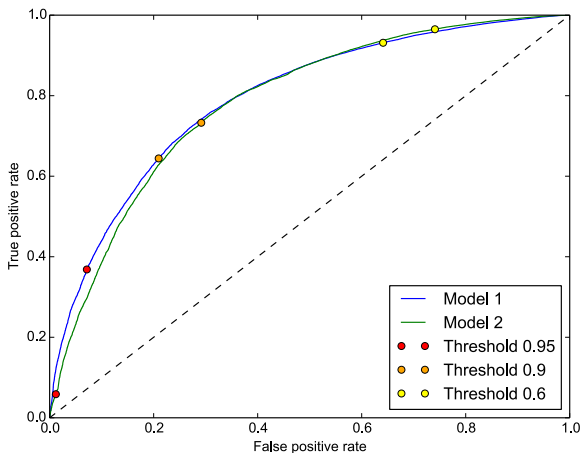
*skewed distribution of classes – hard interpretation
(always use baselines)*

Receiver Operating Characteristic

- to use precision, recall, we need classification into two classes
- probabilistic predictors: value $\in [0, 1]$
- fixed threshold $\Rightarrow$ classification
- what threshold to use? (0.5?)
- evaluate performance over different threshold $\Rightarrow$ Receiver Operating Characteristic (ROC)
- metrics: area under curve (AUC)

AUC used in many domains, sometimes overused

Receiver Operating Characteristic



Metrics for Evaluation of Student Models

Averaging Issues

(relevant for all metrics)

- ratings not distributed uniformly across users/items
- averaging:
 - global
 - per user?
 - per item?
- choice of averaging can significantly influence results
- suitable choice of approach depends on application

Measuring Predictive Performance of User Models: The Details Matter

Ranking

- typical output of RS: **ordered** list of items
- swap on the first place matters more than swap on the 10th place
- ranking metrics – extensions of precision/recall

Ranking Metrics

- Spearman correlation coefficient
- half-life utility
- liftindex
- discounted cumulative gain
- average precision

specific examples for a case study later

Metrics

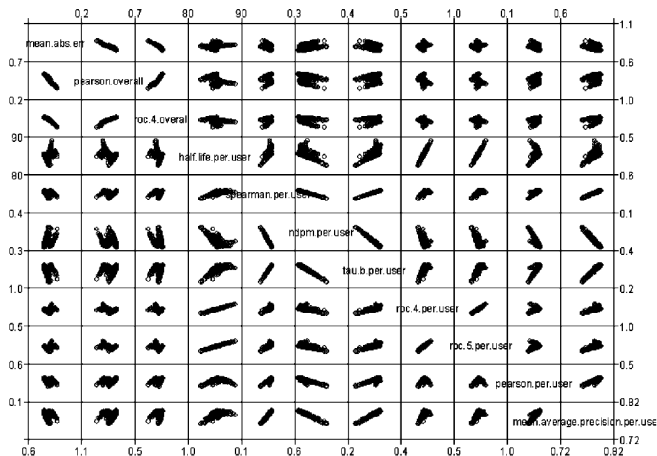
- which metric should we use in evaluation?
- does it matter?

Metrics

- which metric should we use in evaluation?
- does it matter?
- it depends...
- my advice: use RMSE as the basic metric

Metrics for Evaluation of Student Models

Accuracy Metrics – Comparison



Evaluating collaborative filtering recommender systems, Herlocker et al., 2004

Beyond Accuracy of Predictions

harder to measure (user studies may be required) $\Rightarrow$ less used
(but not less important)

- coverage
- confidence
- novelty, serendipity
- diversity
- utility
- robustness

Coverage

- What percentage of items can the recommender form predictions for?
- consider systems X and Y:
 - X provides better accuracy than Y
 - X recommends only subset of “easy-to-recommend” items
- one of RecSys aims: exploit “long tail”

Novelty, Serendipity

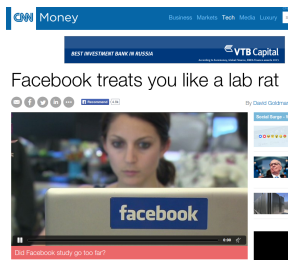
- it is not that difficult to achieve good accuracy on common items
- valuable feature: novelty, serendipity
- serendipity $\sim$ deviation from “natural” prediction
 - successful baseline predictor P
 - serendipity – good, but deemed unlikely by P

Diversity

- often we want diverse results
- example: holiday packages
 - bad: 5 packages from the same resort
 - good: 5 packages from different resorts
- measure of diversity – distance of results from each other
- precision-diversity curve

Online Experiments

- randomized control trial
- AB testing

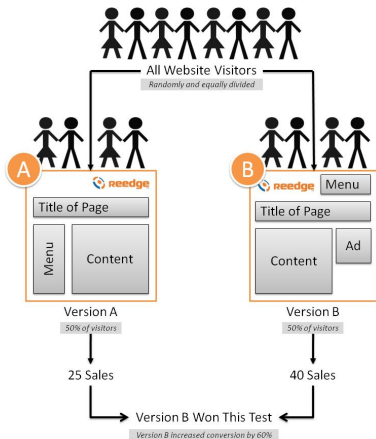


AB Testing

- what is AB testing?
- what is a typical use case?

AB Testing

A/B Testing



Online Experiments – Comparisons

we usually compare averages (**means**)

- are data (approximately) normally distributed?
- if not, averages can be misleading
- specifically: presence of outliers → use **median** or **log transform**

Statistics Reminder

- statistical hypothesis testing
Is my new version really better?
- t-test, ANOVA, significance, p-value
Do I have enough data? Is the observed difference “real” or just due to random fluctuations?
- error bars
How “precise” are obtained estimates?

note: RecSys – very good opportunity to practice statistics

Error Bars

Recommended article: Error bars in experimental biology
(Cumming, Fidler, Vaux)

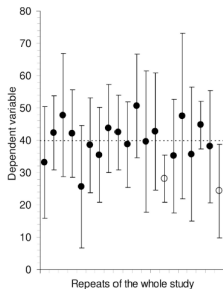


Table I. Common error bars

Error bar	Type	Description	Formula
Range	Descriptive	Amount of spread between the extremes of the data	Highest data point minus the lowest
Standard deviation (SD)	Descriptive	Typical or (roughly speaking) average difference between the data points and their mean	$SD = \sqrt{\frac{\sum (X - M)^2}{n - 1}}$
Standard error (SE)	Inferential	A measure of how variable the mean will be, if you repeat the whole study many times	$SE = SD/\sqrt{n}$
Confidence interval (CI), usually 95% CI	Inferential	A range of values you can be 95% confident contains the true mean	$M \pm t_{[n-1]} \times SE$, where $t_{[n-1]}$ is a critical value of t . If n is 10 or more, the 95% CI is approximately $M \pm 2 \times SE$.

Warning

What you should never do:

report mean value with precision up to 10 decimal places (just because that is the way your program printed the computed value)

Rather:

present only “meaningful” values, report “uncertainty” of your values

Practical Advice

Recommended:

- author Ron Kohavi
- paper *Seven rules of thumb for web site experimenters*
- lecture *Online Controlled Experiments: Lessons from Running A/B/n Tests for 12 Years*

https://www.youtube.com/watch?v=qtboCGd_hTA

context: mainly search engines (but highly relevant for evaluation of recommender systems)

Seven Rules of Thumb for Web Site Experimenters

- ① Small changes can have a big impact to key metrics
- ② Changes rarely have a big positive impact to key metrics
- ③ Your mileage will vary
- ④ Speed matters a lot
- ⑤ Reducing abandonment is hard, shifting clicks is easy
- ⑥ Avoid complex designs: iterate
- ⑦ Have enough users

Number of Users and Detectable Differences

How many users do I need for a meaningful AB experiment?

- hundreds of users – significantly different versions of the system
- tens of thousands of users – different parametrizations of one algorithm
- millions of user – “shades of blue”

Comparing Algorithms Without AB Test

meaningful comparison can be achieved even without splitting users

example:

- two recommendation algorithms A, B
- each picks 3 items
- user is presented with all 6 items (in interleaved order)
- which items users choose more often?

Basic evaluation: this type of comparison, “on ourselves”, compare to “random recommendations”

Simulated Experiments

- simulate data according to a chosen model of users
- add some noise
- advantages:
 - known “ground truth”
 - simple, cheap, fast
 - very useful for testing implementation (bugs in models)
 - insight into behaviour, sensitivity analysis
- disadvantage: results are just consequence of used assumptions

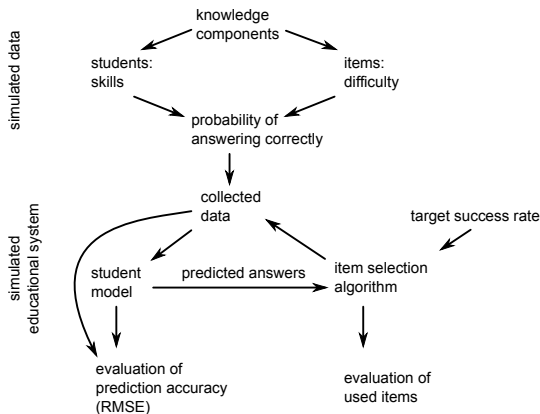
Simple Simulated Setting: Personas

- very simple “manual” simulation
- artificial users (“personas”) with strong preferences
- clear expectations, for which we can check the behavior of the algorithm
- recipe recommendation setting: vegetarian, strict diet, nut allergy, strong preference for Indian food, ...

Simulated Experiments: Simple Example

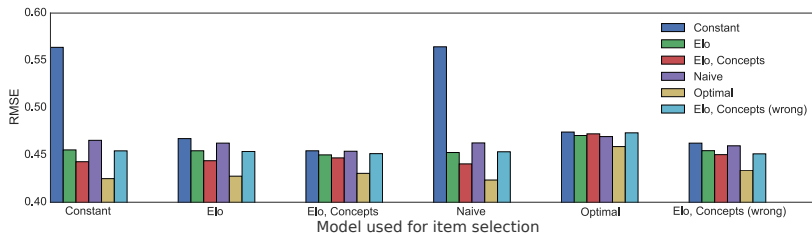
- setting: movies
- simulated users: each user likes some genres (randomly chosen)
- simulated ratings: based on the genre (1 or 4 stars) + random noise
- compute item-item similarity based on ratings
- do the results correspond to genres? how much data needed for convergence?

Simulated Experiments: Realistic Example



Exploring the Role of Small Differences in Predictive Accuracy using Simulated Data

Simulated Experiments: Example

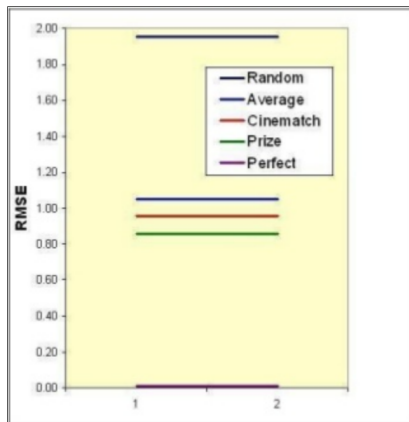


Exploring the Role of Small Differences in Predictive Accuracy using Simulated Data

Interpretation of Results

- what do the numbers mean?
- what do (small) differences mean?
- are they significant?
 - statistically?
 - practically?

Interpretation of Results



Introduction to Recommender Systems, Xavier Amatriain

Magic Barrier

- noise in user ratings / behaviour
- magic barrier – unknown level of prediction accuracy a recommender system can attain
- are we close?
- is further improvement important?

Summary

Proper evaluation is difficult...

- not clear what to measure, how
- things we care about are hard to measure
- many choices that can influence results
 - metrics (RMSE, AUC, ranking...) and their details (thresholds, normalization, averaging...)
 - experimental settings
- it is easy to cheat (unintentionally), overfit

specific examples (case studies) in next lectures

Evaluation and Projects

What kind of evaluation is relevant?

- offline experiments, historical data
- online experiments (AB testing)
- simulated data

How will you perform the evaluation?